Practical : 6

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Problem Statement: Heap Sort -- CO1, CO2, CO4

Design and implement the Heap Sort algorithm to efficiently sort an array of integers in ascending order. The implementation should be optimized for time and space complexity and should clearly demonstrate the working principles of heap data structures (min-heap or maxheap as applicable)

Code

```
# Heap Sort using Max Heap


def heapify(arr, n, i):
    largest = i left
    = 2 * i + 1
    right = 2 * i +
    2


    # Check left child if left < n and
    arr[left] > arr[largest]:


    largest = left


    # Check right child if right < n and
    arr[right] > arr[largest]: largest = right


    # Swap and heapify if needed
    if largest != i:
        arr[i], arr[largest] = arr[largest], arr[i]
```

```
        heapify(arr, n, largest)

def heapSort(arr): n =

len(arr)

    # Build Max Heap
    for i in range(n // 2 - 1, -1, -1):
        heapify(arr, n, i)

    # Sort the array for i in
    range(n - 1, 0, -1):
        arr[0], arr[i] = arr[i], arr[0]
        heapify(arr, i, 0)


# Main Program n = int(input("Enter
number of elements: "))

arr = []

for i in range(n): x =
    int(input("Enter element: "))
    arr.append(x)

print("\nOriginal Array:")
```

```
print(arr)

heapSort(arr)

print("\nSorted Array:")
print(arr)
```

Output

Input

```
Enter number of elements: 5
Enter element: 40
Enter element: 10
Enter element: 30
Enter element: 50
Enter element: 20
```

Output

Original Array:

[40, 10, 30, 50, 20]

Sorted Array:

[10, 20, 30, 40, 50]